

## SUMMARY (narrative by template; all figures from the deterministic engine)

The request for Rs 90,000 over 36 months is referred for manual review under the default policy (v1.0). Verified monthly income is Rs 39,150 against existing obligations of Rs 2,500; the proposed EMI of Rs 3,076 gives a FOIR of 14.2% (limit 50.0%) and an LTV of 75.0% (limit 80.0%). Risk band: MEDIUM. Reason codes: NAME\_MISMATCH.

### For the borrower:

Ravi Kumar, your application for Rs 90,000 needs a manual check before a decision. Some details in your documents did not fully agree, so an underwriter will review them with you.

### Key drivers:

- FOIR 14.2% vs limit 50.0%
- LTV 75.0% vs limit 80.0%
- Verified income Rs 39,150 (volatility 4.1%)
- NAME\_MISMATCH: Full name differs across documents

### Next steps:

- Underwriter to resolve the reconciliation flags with the borrower.
- Re-run the workflow once corrected documents are uploaded.

## 1. DECISION

Decision: REFERRED Confidence: 0.65

Reason codes: NAME\_MISMATCH

- [warning] NAME\_MISMATCH: Full name differs across documents

## 1b. WHY THIS DECISION (policy margins + SHAP; words by template)

Ravi Kumar's loan is referred to a human underwriter because Full name differs across documents.

No hard limit was breached, so an underwriter can still approve it after resolving the flags.

The model's risk band is MEDIUM (score 0.4111 on a 0-1 scale where 0 is safest; a typical synthetic borrower scores 0.437336).

What pushes risk up: identity mismatch flag (yes vs a typical 7% of borrowers, +0.140); ltv (loan / asset value) (75.0% vs a typical 68.1%, +0.012); repayment history score (70.0% vs a typical 71.7%, +0.003).

What pulls risk down: foir (obligations / income) (14.2% vs a typical 41.1%, -0.081); emi-to-income ratio (7.9% vs a typical 34.4%, -0.040); employment stability (95.9% vs a typical 71.8%, -0.024).

The risk score is a secondary signal; the policy rules above decide the outcome.

### Mathematics:

PASS FOIR 14.2% vs limit 50.0% (-35.8 pp under)

PASS LTV 75.0% vs limit 80.0% (-5.0 pp under)

PASS age 29 vs minimum 21 (+8 years)

PASS verified monthly income Rs 39,150

risk score 0.411123 = base 0.437336 + sum SHAP -0.026213 (method additive\_exact, additivity error 0.0)

phi(identity\_mismatch\_flag) = +0.1400 [value yes vs baseline 7% of borrowers]

phi(foir) = -0.0806 [value 14.2% vs baseline 41.1%]

phi(emi\_to\_income) = -0.0398 [value 7.9% vs baseline 34.4%]

phi(employment\_stability) = -0.0241 [value 95.9% vs baseline 71.8%]

phi(income\_volatility) = -0.0183 [value 4.1% vs baseline 22.4%]

## 2. BORROWER SUMMARY (declared facts)

ref\_id: REF-APP-IDENTITY-006

full\_name: Ravi Kumar

age: 29

location\_tier: tier\_2

employment\_type: gig\_platform

declared\_monthly\_income: 39,000.00

loan\_request.amount: 90,000.00

loan\_request.down\_payment: 30,000.00

loan\_request.tenure\_months: 36

loan\_request.vehicle\_price: 120,000.00

### 3. EXTRACTED FACTS (evidence with citations)

full\_name = Ravi Kumar conf 0.95 [application\_form p.None] auto  
monthly\_income = 39,000.00 conf 0.95 [application\_form p.None] auto  
loan\_amount = 90,000.00 conf 0.95 [application\_form p.None] auto  
vehicle\_price = 120,000.00 conf 0.95 [application\_form p.None] auto  
full\_name = Ravi Kumaar conf 0.85 [kyc.txt p.1] auto  
identity = XXXX-XXXX-4321 conf 0.9 [kyc.txt p.1] auto  
address = 14 MG Road, Pune 411001 conf 0.85 [kyc.txt p.1] auto  
monthly\_income = 38,900.00 conf 0.92 [bank\_statement.txt p.1] auto  
bank\_credits = [38500.0, 41000.0, 36200.0, 39800.0, 37900.0, 40100.0] conf 0.9 [bank\_statement.txt p.1] auto  
existing\_monthly\_obligations = 2,500.00 conf 0.92 [bank\_statement.txt p.1] auto  
address = 14 MG Road, Pune 411001 conf 0.85 [bank\_statement.txt p.1] auto  
vehicle\_price = 120,000.00 conf 0.92 [dealer\_invoice.txt p.1] auto  
loan\_amount = 90,000.00 conf 0.92 [dealer\_invoice.txt p.1] auto  
platform\_income = 39,000.00 conf 0.92 [platform\_earnings.txt p.1] auto  
monthly\_income = 39,000.00 conf 0.92 [platform\_earnings.txt p.1] auto

### 4. CALCULATED VALUES (deterministic engine, formula v1)

verified\_income: 39,150.00  
existing\_obligations: 2,500.00  
emi: 3,075.99  
foir: 0.1424  
ltv: 0.75  
income\_volatility: 0.0408  
income\_stability: 0.9592  
loan\_to\_income: 0.1916  
emi\_to\_income: 0.0786  
input.annual\_rate: 14.00  
input.existing\_monthly\_obligations: 2,500.00  
input.loan\_amount: 90,000.00  
input.monthly\_incomes: [38500.0, 41000.0, 36200.0, 39800.0, 37900.0, 40100.0]  
input.tenure\_months: 36  
input.vehicle\_price: 120,000.00

### 5. RISK SIGNAL (secondary, never decides)

score 0.4111 band MEDIUM model xgb-v1/fallback

### 6. AI REASONING

strengths: FOIR 14% is comfortably serviceable  
strengths: LTV 75% leaves borrower equity cushion  
risks: NAME\_MISMATCH: Full name differs across documents  
risks: INCOME\_MISMATCH: Monthly income differs across documents by 0.3%  
reasoning: Deterministic FOIR 14.24%, LTV 75.00%; XGBoost band MEDIUM (score 0.4111).

### 7. POLICY

profile default version 1.0  
limit.foir\_max: 0.5  
limit.ltv\_max: 0.8  
limit.max\_tenure\_months: 60  
limit.min\_age: 21

### 8. HUMAN OVERRIDES

none

SYNTHETIC DATA - demo only. Financial numbers computed deterministically; LLM never sets numbers.